Computing the seasonal PM2.5 contrast

How does processing time change as the workload grows from 8 to 8,000 rows?

Method and workload

Eight observations from two sensors cover one January day and one June day. The heating-season mean exceeds the non-heating mean by 75.9 micrograms/m³. Repeating the rows gives four workloads with the same contrast. Each implementation ran five warm-up trials followed by 30 evaluation trials per size.

In-process latency by implementation

Rows	Row loop (ms)	Pandas (ms)	Speedup
8	0.587 ± 0.206	0.510 ± 0.252	1.15x
80	2.754 ± 0.186	0.265 ± 0.016	10.39x
800	26.862 ± 3.531	0.349 ± 0.164	77.05x
8,000	320.405 ± 90.569	0.555 ± 0.172	577.78x

Values are mean ± sample SD. Speedup is the ratio of unrounded means.

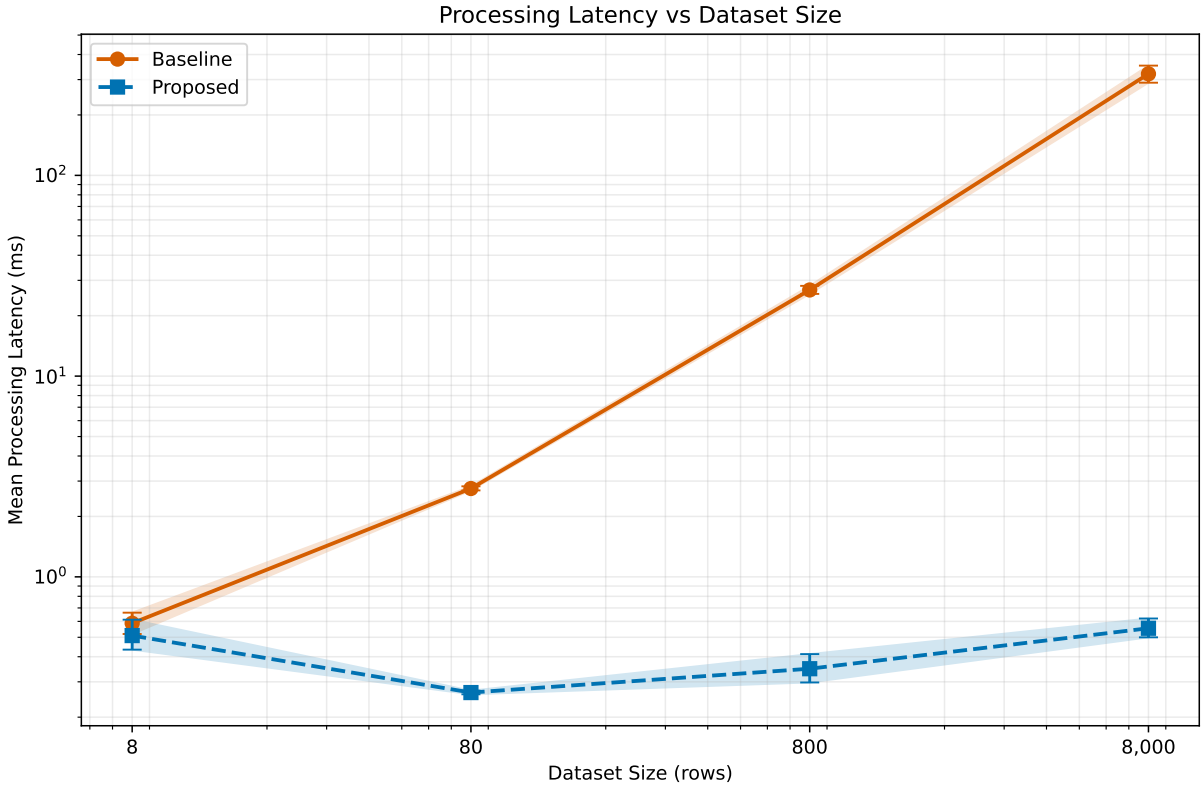

Figure 1. Logarithmic axes show the widening time gap. Bars and shaded bands show 95% bootstrap intervals for the mean. All 240 evaluation attempts succeeded.

Process time and interpretation

A separate experiment timed the complete Python program on the eight-row sample.

30 successful trials after five warm-up runs

A fresh process started for each trial. The timer includes imports, file reads and logging.

Mean 616.80 ms | SD 53.49 ms | Median 609.67 ms

P95 717.09 ms | P99 740.53 ms | Observed errors 0/30

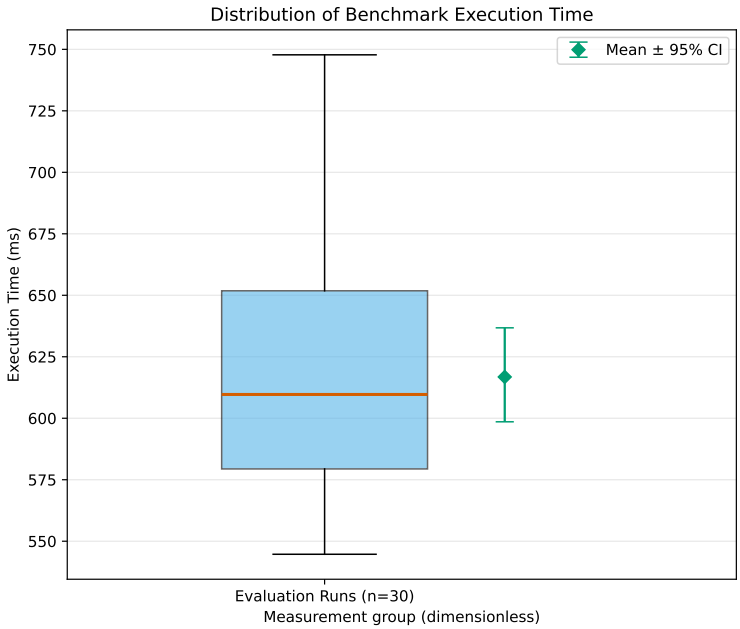

Figure 2. The box spans the middle half of individual runtimes. The diamond marks the mean and its 95% interval (598.56 to 636.77 ms, 10,000 resamples, seed 42).

What the measurements support

The Pandas implementation took less time on average at every size in this run. At eight rows, the mean intervals overlap. At 8,000 rows the gap is large, but both methods ran in fixed order on one machine. Order effects and the original hardware cannot be separated from the timings. P99 is especially sensitive to individual trials.

Only eight distinct environmental observations enter the seasonal contrast. All four heating observations exceed the project spike threshold of 35 micrograms/m³, and none of the four summer observations do. These data cannot identify a heating-by-calm interaction. The repository defines a logistic model and log loss for future observations.

Open the tagged project, notebook and raw measurements on GitHub